AN INJECTIVE ABELIAN SHEAF IS INJECTIVE AS A PRESHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the adjunction and the consequence are both inside the counted span.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Injective abelian sheaf and category of abelian presheaves. Same chapter, the two categories involved and the forgetful functor between them. Supplies the objects compared and the functor tested.
- Adjunction between sheafification and the forgetful functor, and exactness of sheafification. Same chapter, the adjunction whose left adjoint is sheafification, together with exactness of sheafification. Supplies the formal argument transferring injectivity.

Source reference. Original lemma 015Z: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1325–1325. Navigation pointer invoked at proof line 1325; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03CM: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1329–1329. Navigation pointer invoked at proof line 1329; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. An injective abelian sheaf is also injective as an object in the category $PAb(\mathcal{C})$.*

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Proof. Apply Homology, Lemma 015Z to the categories $\mathcal{A} = Ab(\mathcal{C})$, $\mathcal{B} = PAb(\mathcal{C})$, the inclusion functor and sheafification. (See Modules on Sites, Section 03CM to see that all assumptions of the lemma are satisfied.) □